

JASPER COUNTY AP, TX, USA

WMO: 720296

Lat: 30.886N

Lon: 94.035W

Elev: 65

StdP: 100.55

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53945

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.1	-0.5	-8.1	1.9	4.7	-5.6	2.4	6.2	8.3	12.7	7.5	12.7	0.8	10	0.343

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.2	36.4	24.4	35.1	24.6	34.0	24.8	26.9	31.3	26.6	31.1	26.2	30.8	2.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.6	28.7	25.5	20.9	28.4	25.1	20.4	28.2	85.1	31.5	83.4	31.1	81.7	30.8	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	5.8	5.1	DB	-5.1	37.4	1.9	1.8	-6.5	38.7	-7.6	39.7	-8.7	40.7	-10.1	42.0	(4)
(5)			WB	-5.5	28.0	2.0	1.0	-6.9	28.7	-8.1	29.3	-9.2	29.9	-10.6	30.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.9	10.3	12.6	16.4	19.5	23.6	26.9	27.6	28.3	25.8	20.8	15.1	11.6	(6)	
	DBStd	7.53	5.32	5.63	5.00	3.93	3.31	2.01	1.78	1.98	2.52	4.33	5.15	5.40	(7)	
	HDD10.0	161	62	33	8	1	0	0	0	0	0	0	12	44	(8)	
	HDD18.3	930	256	174	96	32	4	0	0	0	0	26	122	219	(9)	
	CDD10.0	3780	71	106	208	286	420	506	547	568	475	333	167	94	(10)	
	CDD18.3	1508	6	13	38	67	166	256	288	310	225	101	27	12	(11)	
	CDH23.3	15044	14	58	224	560	1581	2705	3123	3503	2149	928	162	36	(12)	
(13)	CDH26.7	6613	0	6	29	112	591	1279	1496	1792	977	306	24	1	(13)	
(14)	Wind	WSAvg	1.7	2.1	2.3	2.3	2.2	1.8	1.4	1.2	1.3	1.5	1.8	2.0	(14)	
(15) Precipitation	PrecAvg	1392	130	105	107	109	102	124	98	101	110	127	132	139	(15)	
	PrecMax	1968	277	271	216	256	247	330	199	201	291	535	338	278	(16)	
	PrecMin	827	26	13	20	13	0	33	24	0	41	0	13	44	(17)	
	PrecStd	273	75	72	55	61	63	71	48	55	67	127	86	60	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.0	27.6	29.0	31.0	34.0	37.2	36.8	38.6	35.8	33.0	29.4	26.4	(19)	
		MCWB	19.0	20.5	20.1	21.3	23.1	23.6	24.3	24.1	24.0	23.2	22.5	21.4	(20)	
	2%	DB	23.1	25.2	27.5	29.2	32.6	35.3	35.4	37.0	34.7	31.4	27.1	24.2	(21)	
		MCWB	18.2	19.3	19.7	20.6	23.3	24.2	24.7	24.6	24.0	23.0	21.3	20.7	(22)	
	5%	DB	21.7	23.2	26.1	27.9	31.4	33.8	34.1	35.4	33.3	30.0	25.1	22.8	(23)	
		MCWB	18.2	19.2	19.4	20.3	23.5	24.9	25.2	25.0	24.0	22.6	20.1	20.2	(24)	
	10%	DB	19.9	21.7	24.1	26.7	30.2	32.6	33.1	34.0	32.0	28.3	23.5	21.2	(25)	
		MCWB	17.0	18.5	18.5	20.0	23.0	24.9	25.4	25.1	23.8	21.9	19.7	18.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.1	21.9	22.3	24.3	26.1	27.3	27.5	27.8	26.9	26.5	24.0	22.4	(27)	
		MCDB	22.9	24.6	25.7	27.2	30.2	31.6	31.5	31.7	30.3	30.0	27.5	24.5	(28)	
	2%	WB	20.3	21.2	21.6	23.2	25.2	26.7	26.8	27.1	26.2	25.5	23.0	21.7	(29)	
		MCDB	22.1	23.7	24.8	26.6	29.6	31.1	31.2	31.5	30.6	28.8	25.4	23.5	(30)	
	5%	WB	19.1	20.3	21.0	22.5	24.6	26.2	26.5	26.6	25.7	24.6	21.7	21.0	(31)	
		MCDB	21.0	22.6	23.9	25.9	28.9	30.6	31.1	31.3	30.2	27.8	24.2	22.7	(32)	
	10%	WB	17.6	19.0	20.3	21.8	24.0	25.7	26.1	26.2	25.1	23.6	20.4	19.3	(33)	
		MCDB	19.1	21.2	23.2	25.0	28.2	30.0	30.6	30.9	29.6	26.7	22.8	20.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	11.3	11.9	12.2	11.5	10.8	10.6	11.2	11.5	12.6	12.2	10.8	(35)	
		MCDBR	12.8	11.6	11.6	12.4	12.2	12.6	12.1	13.0	12.8	12.8	12.5	11.3	(36)	
	5% WB	MCWBR	8.4	6.8	5.5	5.5	4.5	4.2	3.9	3.5	4.4	5.7	7.2	7.8	(37)	
		MCWBR	10.0	9.7	9.7	9.8	10.5	10.6	10.8	10.9	10.9	10.1	10.3	9.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.329	0.349	0.369	0.417	0.456	0.488	0.492	0.483	0.445	0.374	0.364	0.341	(40)	
		taud	2.533	2.484	2.423	2.295	2.223	2.171	2.200	2.225	2.296	2.474	2.451	2.503	(41)	
	Ebn at Noon		902	913	914	878	841	810	805	809	828	873	852	867	(42)	
		Edh at Noon	86	99	112	132	143	150	146	140	126	99	93	85	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.80	3.12	4.36	5.31	5.88	6.07	6.03	5.72	4.98	4.23	3.16	2.43	(44)	
	RadStd		0.26	0.54	0.39	0.35	0.42	0.46	0.47	0.49	0.46	0.57	0.33	0.29	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47)	Regional (38 neighbors)	+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+105	+83	(47)	

Nomenclature: See separate page